PRACTICAL – 6

Rendering External JSON Data using Fetch API, Search and Filter

Aim:-create events in events.json and make API and fatch that API in Events.html so that can see in Events section in Website.

Technologies Used:-

- HTML5
- CSS3
- Json

Implementation:-

Code:

Events. Json

```
[
  {
    "name": "AI Workshop",
    "category": "Workshop",
    "date": "20-08-2026"
  },
  {
    "name": "Hackathon",
    "category": "Competition",
    "date": "22-08-2026"
  },
```

```
{
    "name": "Flutter Workshop",
    "category": "Workshop",
    "date": "25-08-2026"
},
{
    "name": "Coding Contest",
    "category": "Competition",
    "date": "28-08-2026"
},
{
    "name": "Web Seminar",
    "category": "Seminar",
    "date": "01-09-2026"
},
{
    "name": "Python Workshop",
    "category": "Workshop",
    "date": "03-09-2026"
},
{
    "name": "Tech Talk",
    "category": "Seminar",
    "date": "05-09-2026"
},
{
```

```
    "name": "Java Contest",
    "category": "Competition",
    "date": "08-09-2026"
  },
  {
    "name": "Cyber Security Workshop",
    "category": "Workshop",
    "date": "10-09-2026"
  },
  {
    "name": "AI Seminar",
    "category": "Seminar",
    "date": "12-09-2026"
  },
  {
    "name": "Code Hackathon",
    "category": "Competition",
    "date": "15-09-2026"
  },
  {
    "name": "Cloud Workshop",
    "category": "Workshop",
    "date": "18-09-2026"
  },
  {
    "name": "Data Science Seminar",
```

```
        "category": "Seminar",
        "date": "20-09-2026"
    },
    {
        "name": "Web Design Contest",
        "category": "Competition",
        "date": "22-09-2026"
    },
    {
        "name": "JavaScript Workshop",
        "category": "Workshop",
        "date": "25-09-2026"
    }
]
```

Events.html

```
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" href="style.css">
    <title>College Events</title>
    <style>
        body {
            font-family: Arial;
            margin: 30px;
```

```
        }
        h1 {
            text-align: center;
        }
        input, select, button {
            padding: 8px;
            margin: 5px;
        }
        .event {
            border: 1px solid black;
            padding: 10px;
            margin: 10px;
        }
    </style>
</head>
<body>
    <h1>College Events</h1>
    <input type="text" id="search" placeholder="Search event">
    <select id="category">
        <option value="All">All</option>
        <option value="Workshop">Workshop</option>
        <option value="Seminar">Seminar</option>
        <option value="Competition">Competition</option>
    </select>
    <button onclick="sortEvents()">
        Sort by Name
```

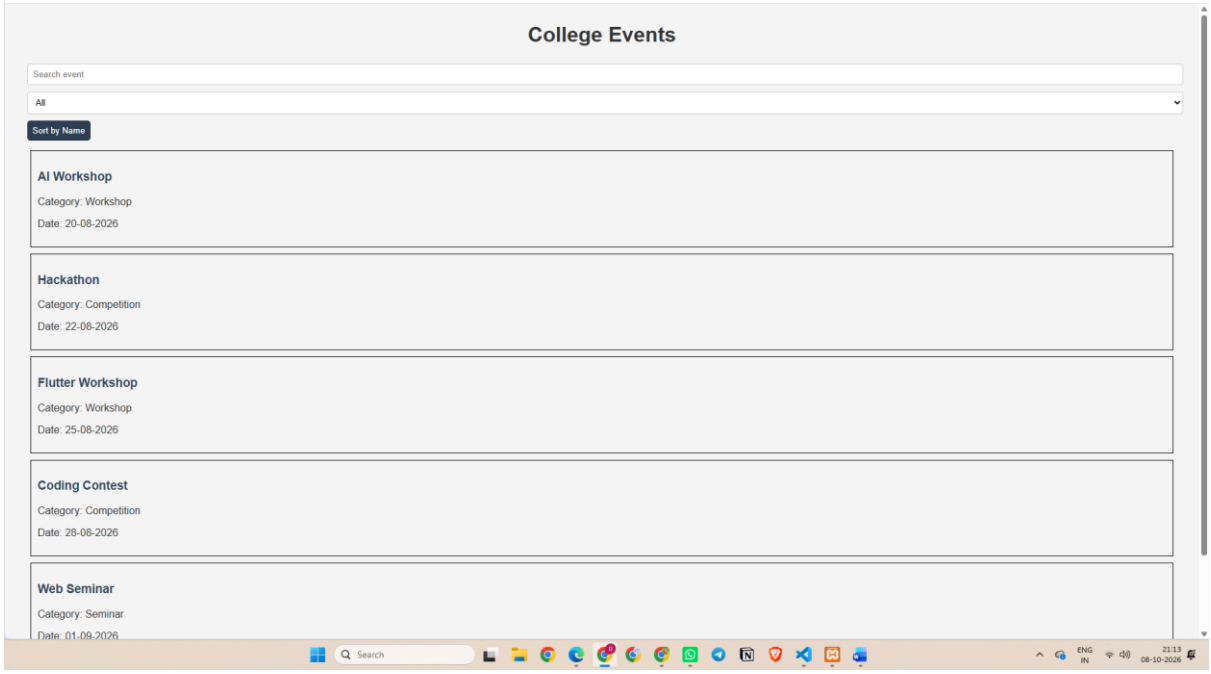
```
        </button>
        <div id="output">Loading...</div>
        <br>
        <button onclick="previousPage()">
            Previous
        </button>
        <button onclick="nextPage()">
            Next
        </button>
        <script>
            let events = [];
            let page = 1;
            let perPage = 5;
            fetch("events.json")
                .then(response => response.json())
                .then(data => {
                    events = data;
                    displayEvents();
                })
                .catch(error => {
                    document.getElementById("output").innerHTML =
                        "Error loading data";
                });
            function displayEvents() {
                let searchText = document.getElementById("search")
.value.toLowerCase();
```

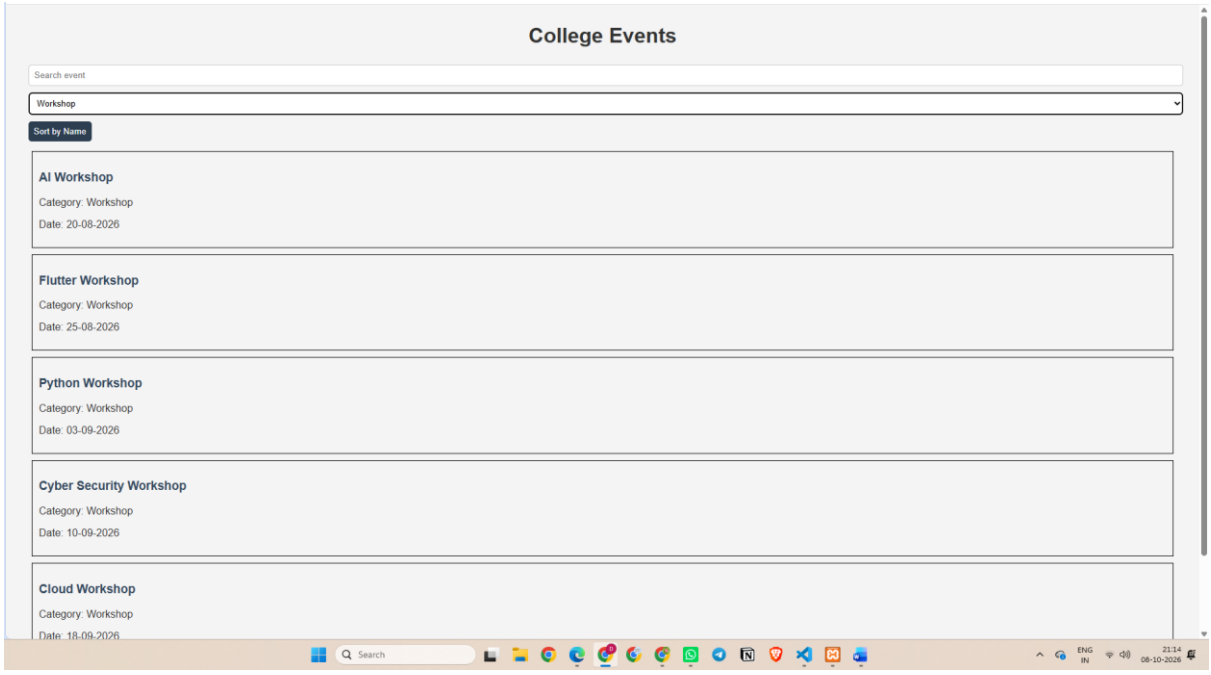
```
        let category =document.getElementById("category").value;
        // Search
        let result = events.filter(function(event) {
            return event.name .toLowerCase() .includes(searchText);
        });
        if (category != "All") {
            result = result.filter(function(event) {
                return event.category == category;
            });
        }
        let start = (page - 1) * perPage;
        let end = start + perPage;
        let pageData = result.slice(start, end);
        let output = "";
        pageData.forEach(function(event) {
            output +="<div class='event'>" + "<h3>" + event.name +
"</h3>" +
                "<p>Category: " + event.category + "</p>" +
                "<p>Date: " + event.date + "</p>" +
                "</div>";
        });
        document.getElementById("output")
            .innerHTML = output;
    }
    document.getElementById("search")
        .addEventListener("input", function() {
```

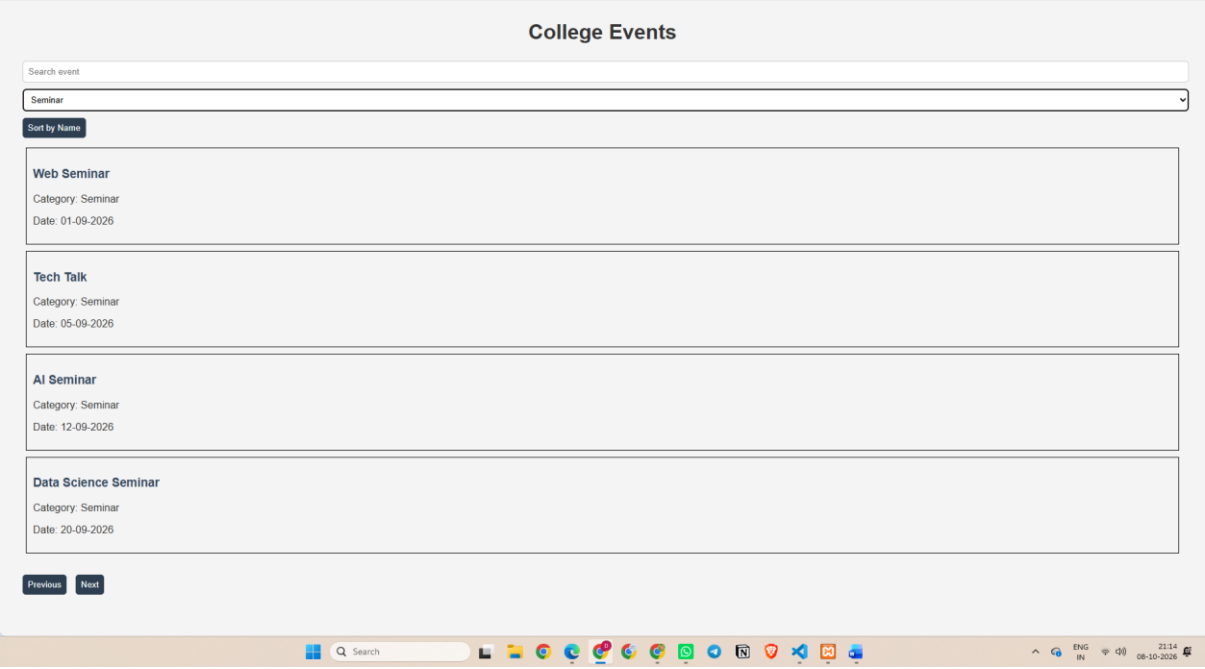
```
            page = 1;
            displayEvents();
        });
        document.getElementById("category").addEventListener("change",
function() {
            page = 1;
            displayEvents();
        });
        function sortEvents() {
            events.sort(function(a, b) {
                return a.name.localeCompare(b.name);
            });
            displayEvents();
        }
        function nextPage() {
            page++;
            displayEvents();
        }
        function previousPage() {
            if (page > 1) {
                page--;
                displayEvents();
            }
        }
    </script>
</body>
```

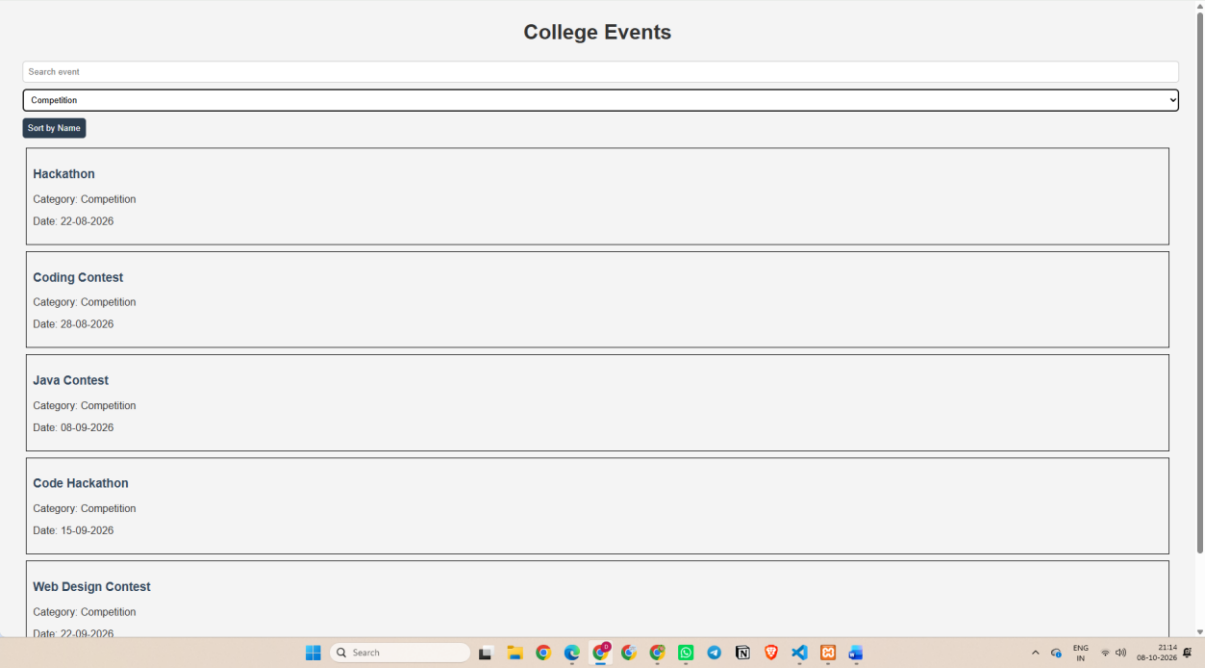
```
</html>
```

OUTPUT: -


College Events
Search event
All
Sort by Name
AI Workshop
Category: Workshop
Date: 20-08-2026
Hackathon
Category: Competition
Date: 22-08-2026
Flutter Workshop
Category: Workshop
Date: 25-08-2026
Coding Contest
Category: Competition
Date: 28-08-2026
Web Seminar
Category: Seminar


College Events
Search event
Workshop
Sort by Name
AI Workshop
Category: Workshop
Date: 20-08-2026
Flutter Workshop
Category: Workshop
Date: 25-08-2026
Python Workshop
Category: Workshop
Date: 03-09-2026
Cyber Security Workshop
Category: Workshop
Date: 10-09-2026
Cloud Workshop
Category: Workshop


College Events
Search event
Seminar
Sort by Name
Web Seminar
Category: Seminar
Date: 01-09-2026
Tech Talk
Category: Seminar
Date: 05-09-2026
AI Seminar
Category: Seminar
Date: 12-09-2026
Data Science Seminar
Category: Seminar
Date: 20-09-2026
Previous
Next


College Events
Search event
Competition
Sort by Name
Hackathon
Category: Competition
Date: 22-08-2026
Coding Contest
Category: Competition
Date: 28-08-2026
Java Contest
Category: Competition
Date: 08-09-2026
Code Hackathon
Category: Competition
Date: 15-09-2026
Web Design Contest
Category: Competition